

A SHORT VISUAL TOUR

From image processing to GANs

Four stages of computer vision, each with a real entertainment industry example



1

1970s–90s

Image processing & feature extraction

Pixel-level filtering and hand-built descriptors — edges, corners, motion markers — laid the groundwork for digital visual effects, before any of it involved machine learning.

REAL USE CASE

Optical compositing built the layered lightsaber and space-battle shots in the original Star Wars. Marker-based motion capture later drove early performance-driven CGI characters.

Classical ML & deep learning (CNNs)

Classifiers first ran on hand-built features. Then convolutional networks took over, learning hierarchical visual features straight from raw pixels.

REAL USE CASE

CNN-based recommendation engines reshaped how Netflix and YouTube surface content to viewers, while neural style-transfer apps like Prisma brought the technique to a mass audience.

3

2013–16

Generative models (VAEs)

Variational autoencoders learned to model and reconstruct entire data distributions — not just classify inputs, but generate plausible new ones.

REAL USE CASE

AI-assisted colorization and restoration techniques, in the same spirit as what powered Peter Jackson's WWI documentary *They Shall Not Grow Old*, brought archival footage back to life.

4

2014–PRESENT

GANs

A generator and a discriminator trained adversarially, pushing synthetic media all the way to photorealistic quality.

REAL USE CASE

Deepfake-based actor de-aging — young Harrison Ford in Indiana Jones and the Dial of Destiny — plus StyleGAN-generated virtual influencers and game characters.